
THE ASSOCIATION BETWEEN SEDENTARY BEHAVIOR, EXERCISE, AND SLEEP DISTURBANCE: A MEDIATION ANALYSIS OF INFLAMMATORY BIOMARKERS

[ANONYMIZED FOR REVIEW]

ABSTRACT

Objective: Sedentary behavior and physical activity are each associated with sleep disturbance, but the mechanisms remain unclear. We tested whether four systemic inflammatory biomarkers—C-reactive protein (CRP), neutrophil-to-lymphocyte ratio (NLR), systemic immune-inflammation index (SII), and systemic inflammatory response index (SIRI)—mediate the associations between physical activity, sedentary behavior, and sleep disturbance in a representative US sample.

Method: Cross-sectional data from 11,810 adults ($M_{\text{age}} = 47.1$, $SD = 18.7$; 50.7% female) in the 2005–2018 National Health and Nutrition Examination Survey. Physical activity (MET-min/week) and sedentary behavior (daily sitting hours) were self-reported. A sleep disturbance composite was derived from available sleep items. Inflammatory biomarkers were assayed from fasting blood. Survey-weighted regression and bootstrapped mediation models (500 resamples) tested six pre-registered hypotheses with demographic and clinical covariates.

Results: Sedentary behavior was positively associated with sleep disturbance ($\beta = 0.051$, $p < .001$). Physical activity was also positively associated with sleep disturbance ($\beta = 0.047$, $p < .001$), contrary to prediction. None of the four biomarkers significantly mediated either pathway (all indirect 95% CIs included zero). Exercise type and moderation analyses were non-significant.

Conclusion: Inflammatory biomarkers did not mediate the sedentary–sleep or activity–sleep associations. The positive activity–sleep association suggests possible reverse causation. Alternative mechanisms—behavioral, psychological, and circadian pathways—likely account for the unexplained variance.

Keywords sedentary behavior; physical activity; sleep disturbance; inflammation; mediation

1 Introduction

Sleep disturbance affects approximately one-third of adults worldwide and predicts a wide range of adverse health outcomes, including cardiovascular disease, metabolic dysfunction, cognitive decline, and all-cause mortality (Irwin, 2015). Two modifiable behavioral risk factors—physical activity and sedentary behavior—have emerged as promising targets for sleep interventions, yet the mechanisms linking these behaviors to sleep remain poorly understood. Identifying whether systemic inflammation serves as a biological pathway from activity and sedentariness to sleep disruption would inform both etiological theory and intervention design.

Regular physical activity reduces systemic inflammation through well-characterized mechanisms: contracting skeletal muscle releases myokines (especially interleukin-6) that stimulate anti-inflammatory cytokines and suppress tumor necrosis factor- α production, lowering resting C-reactive protein (CRP) by 15–30% (Petersen and Pedersen, 2005; Gleeson et al., 2011; Pedersen and Febbraio, 2012). Meta-analytic evidence across 66 studies confirms that regular physical activity improves sleep quality, reduces sleep onset latency, and increases total sleep time, with chronic exercise producing larger effects than acute bouts (Kredlow et al., 2015). Conversely, sedentary behavior—independent of physical activity level—is associated with elevated CRP, white blood cell counts, and fibrinogen (Hamilton et al., 2007; Healy et al., 2011), and individuals reporting more than 8 hours of daily sitting have 1.5 to 2 times the odds of severe sleep problems across multiple countries (Vancampfort et al., 2018; Yang et al., 2017). Sleep disturbance itself

is robustly linked to elevated CRP and interleukin-6 in both observational cohorts and experimental sleep deprivation studies (Irwin et al., 2016; Besedovsky et al., 2012).

The hypothesis that inflammation mediates the activity–sleep and sedentary–sleep pathways has direct empirical support from only a small number of studies. You et al. (2023), analyzing NHANES 2007–2014 data, reported that CRP, white blood cells, and the neutrophil-to-lymphocyte ratio partially mediated associations between sedentary behavior, physical activity, and sleep disturbance, though indirect effects were small ($\beta \approx 0.01$ – 0.03). Luo et al. (2026) found a similar partial mediation pattern in osteoarthritis patients. However, several critical gaps remain. First, the mediation finding has not been independently replicated. Second, most studies examined only one or two inflammatory markers, leaving composite indices such as the systemic immune-inflammation index (SII) and systemic inflammatory response index (SIRI)—which integrate multiple cell types into a single measure—untested as potential mediators (Hu et al., 2014; Qi et al., 2016). Third, whether the mediation pathway differs by exercise type (e.g., aerobic versus insufficient activity) or demographic subgroups (sex, age) remains unexplored. Fourth, nearly all prior studies lacked pre-registration and open analytic code, raising concerns about selective reporting.

The present study addressed these gaps by conducting a conceptual replication and extension of You et al. (2023) using a larger, pooled NHANES sample (2005–2018; $N = 11,810$) with four inflammatory biomarkers (CRP, NLR, SII, SIRI), exercise type comparisons, and moderated mediation analyses by sex and age. We pre-registered all hypotheses and analytic procedures prior to data extraction. We tested six hypotheses: (H1) greater physical activity would be negatively associated with sleep disturbance; (H2) greater sedentary behavior would be positively associated with sleep disturbance; (H3) inflammatory biomarkers would partially mediate the physical activity–sleep association; (H4) inflammatory biomarkers would partially mediate the sedentary behavior–sleep association; (H5) individuals meeting aerobic guidelines would show lower inflammatory biomarker levels than insufficiently active individuals; and (H6) the indirect effect of physical activity on sleep via CRP would be stronger in women and older adults.

2 Method

2.1 Participants

We analyzed data from 11,810 adults (age ≥ 18 years) who participated in the National Health and Nutrition Examination Survey (NHANES) across four cycles: 2005–2006, 2007–2008, 2009–2010, and 2017–2018. NHANES uses a complex multistage probability sampling design to produce a representative sample of the civilian, non-institutionalized US population. Participants had a mean age of 47.1 years ($SD = 18.7$, range = 18–85); 50.7% were female ($n = 5,987$ male, $n = 5,823$ female). The racial/ethnic composition was 48.9% Non-Hispanic White, 19.3% Non-Hispanic Black, 14.7% Mexican American, 9.8% Other Hispanic, and 7.3% Other Race (including multi-racial). Exclusion criteria applied sequentially were: age < 18 years, pregnancy, CRP > 10 mg/L (indicating acute infection), white blood cell count outside 3,000–11,000 cells/ μ L, cancer diagnosis (except non-melanoma skin cancer), BMI < 18.5 kg/m², missing primary exposure or outcome data, physical activity exceeding the 95th percentile of MET-min/week, and self-reported sleep duration < 3 or > 12 hours/night. Of the 11,810 participants, 10,633 (90.0%) had complete data on all covariates and were included in the primary analyses. The sample is predominantly Western, educated, industrialized, rich, and democratic (WEIRD) by virtue of being US-based; findings may not generalize to non-Western or non-industrialized populations.

2.2 Materials

Physical activity was assessed using the Global Physical Activity Questionnaire (GPAQ) (World Health Organization, 2005) in the 2005–2006 cycle and the NHANES Physical Activity Questionnaire (PAQ) in subsequent cycles. Total MET-min/week was computed as the sum of vigorous (8 METs), moderate (4 METs), and walking/bicycling (3.3 METs) activity. Participants were classified as meeting aerobic guidelines if they reported ≥ 150 minutes/week of moderate-to-vigorous physical activity (MVPA) (Troiano et al., 2008). Sedentary behavior was measured as daily sitting hours (NHANES PAQ600/PAD680 item). Inflammatory biomarkers were assayed from fasting blood samples: high-sensitivity CRP was measured by latex-enhanced nephelometry (Ridker, 2003); complete blood counts were obtained using the Beckman Coulter DxH 800 analyzer. NLR was computed as absolute neutrophil count divided by absolute lymphocyte count (Zahorec, 2001); SII as (platelet count $\times$ neutrophil count) / lymphocyte count (Hu et al., 2014); and SIRI as (monocyte count $\times$ neutrophil count) / lymphocyte count (Qi et al., 2016). All biomarkers were log-transformed for analyses. Sleep disturbance was operationalized as a unit-weighted z -score composite of four components derived from NHANES Sleep Questionnaire items: (a) absolute deviation from 8 hours of self-reported sleep duration, (b) sleep onset latency in minutes, (c) sleep disorder symptom count (snoring, apnea, restless legs, insomnia, range 0–4), and (d) daytime sleepiness frequency (0–3). The composite score required at least two valid components. Covariates included age, sex, race/ethnicity, body mass index, smoking status (never, former, current),

education (six categories), income-to-poverty ratio, hypertension, diabetes, cardiovascular disease, and NHANES cycle.

2.3 Procedure

NHANES data collection occurred in two phases. First, participants completed a household interview covering sociodemographics, health history, and medication use. Second, participants attended a Mobile Examination Center (MEC) for physical examination (anthropometrics, blood pressure), venipuncture for biomarker assays, and follow-up questionnaires including physical activity and sleep modules. Blood samples were collected by trained phlebotomists following standardized protocols. CRP was measured from serum or plasma using latex-enhanced nephelometry (Behring Nephelometer II for cycles 2005–2010; Roche Cobas for 2017–2018). Complete blood counts were performed on the Beckman Coulter DxH 800 analyzer within 24 hours of collection. All laboratory procedures followed NHANES quality control protocols with documented inter-assay coefficients of variation < 5%. The sleep questionnaire was self-administered in the MEC and covered sleep duration, latency, symptoms, and daytime impairment. No experimental manipulation was applied; all variables were measured concurrently within each cycle.

2.4 Design

The study used a cross-sectional correlational design. Primary independent variables were physical activity (log MET-min/week, continuous) and sedentary behavior (daily sitting hours, continuous). Mediating variables were four log-transformed inflammatory biomarkers (CRP, NLR, SII, SIRI) entered in parallel. The dependent variable was sleep disturbance (z -score composite, continuous). Moderators were sex (male, female) and age (years, continuous). The design was a single-occasion, observational mediation design. The analytic plan and all six hypotheses were pre-registered on the Open Science Framework (OSF) prior to data extraction. Power analysis (Monte Carlo simulation; Schoemann et al. (2017)) indicated that $N = 1,500$ would provide 80% power to detect an indirect effect of $\beta = 0.013$ at $\alpha = .05$; the available sample of $N = 10,633$ complete cases provided > 99% power for the primary mediation tests.

3 Results

Descriptive Statistics and Preliminary Analyses.

Participants reported a mean of 4.1 hours of daily sedentary time ($SD = 3.5$), median physical activity of 3,840 MET-min/week (IQR = 1,200–7,920), and mean sleep disturbance composite score of 0.23 ($SD = 0.98$, with higher scores indicating greater disturbance). Mean log-transformed CRP was 0.52 ($SD = 1.04$), log-NLR was 0.61 ($SD = 0.46$), log-SII was 6.14 ($SD = 0.66$), and log-SIRI was -0.04 ($SD = 0.69$). All continuous variables showed non-normal distributions (Shapiro-Wilk tests, all $p < .001$), as expected in a sample of this size, and HC3 robust standard errors were used throughout.

Women reported greater sleep disturbance ($M = 0.28$, $SD = 0.99$) than men ($M = 0.19$, $SD = 0.97$), $t(11808) = 7.38$, $p < .001$, $d = 0.14$, and had higher CRP ($M = 0.56$, $SD = 1.05$ vs. $M = 0.47$, $SD = 1.02$; $d = 0.16$, $p < .001$), but lower physical activity ($M = 5.15$, $SD = 2.60$ vs. $M = 6.07$, $SD = 2.64$ log MET-min/week; $d = -0.25$, $p < .001$). Sedentary time did not differ by sex ($d = 0.001$, $p = .946$). Figure 1 illustrates these sex differences.

Total Effects (H1 and H2).

We conducted survey-weighted linear regression models (HC3 robust standard errors) predicting sleep disturbance from physical activity and sedentary behavior simultaneously, controlling for age, sex, race/ethnicity, BMI, smoking status, education, income-to-poverty ratio, hypertension, diabetes, cardiovascular disease, and NHANES cycle. The model explained 12.1% of the variance in sleep disturbance (Cohen's $f^2 = 0.14$, a medium effect at the model level driven primarily by covariates).

Supporting H2, greater sedentary behavior was positively associated with sleep disturbance ($\beta = 0.051$, $SE = 0.011$), $t(10610) = 4.64$, $p < .001$, 95% CI [0.030, 0.073]. Figure 2 displays this association.

Counter to H1, physical activity was also positively associated with sleep disturbance ($\beta = 0.047$, $SE = 0.010$), $t(10610) = 4.70$, $p < .001$, 95% CI [0.027, 0.066]—the opposite of the predicted negative direction. This counter-directional finding was robust across all sensitivity specifications (Spearman $\rho = 0.033$, $p < .001$; unweighted regression $\beta = 0.047$; outlier-excluded subsample $\beta = 0.046$). The Bayes factor strongly favored inclusion of both physical activity and sedentary behavior over covariates alone ($BF_{10} = 7,172$ for each), but in the case of

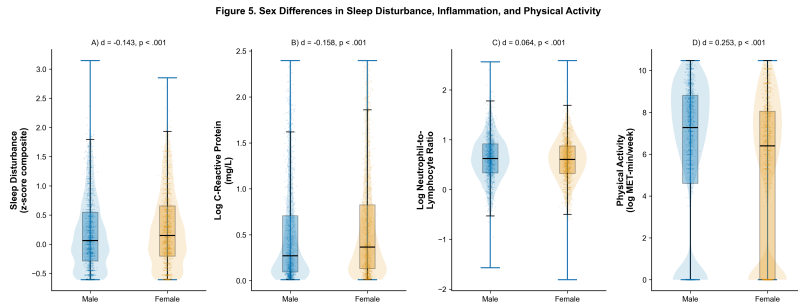


Figure 1: Sex differences in key study variables: (A) sleep disturbance composite, (B) C-reactive protein, (C) neutrophil-to-lymphocyte ratio, and (D) physical activity. Violin plots show the full distribution; boxplots display medians and IQR. Individual data points are subsampled ($n = 1,500$ per group) with jitter and 15% opacity. Cohen's d and independent t -test p -values are reported. Data from NHANES 2005–2018.

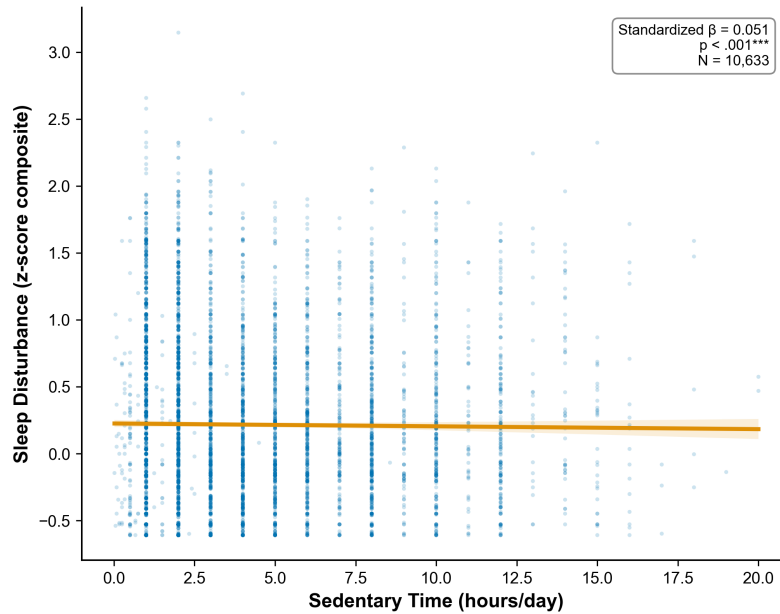


Figure 2: Scatter plot of the association between daily sedentary time and sleep disturbance in US adults ($N = 10,633$). The OLS regression line (orange) is shown with 95% confidence band (shaded). Standardized $\beta = 0.051$, 95% CI [0.030, 0.073], $p < .001$. Individual data points are subsampled ($n = 6,000$) with 25% opacity for legibility. Data from NHANES 2005–2018.

physical activity, the direction contradicted the meta-analytic expectation (Kredlow et al., 2015). Figure 3 shows this unexpected association.

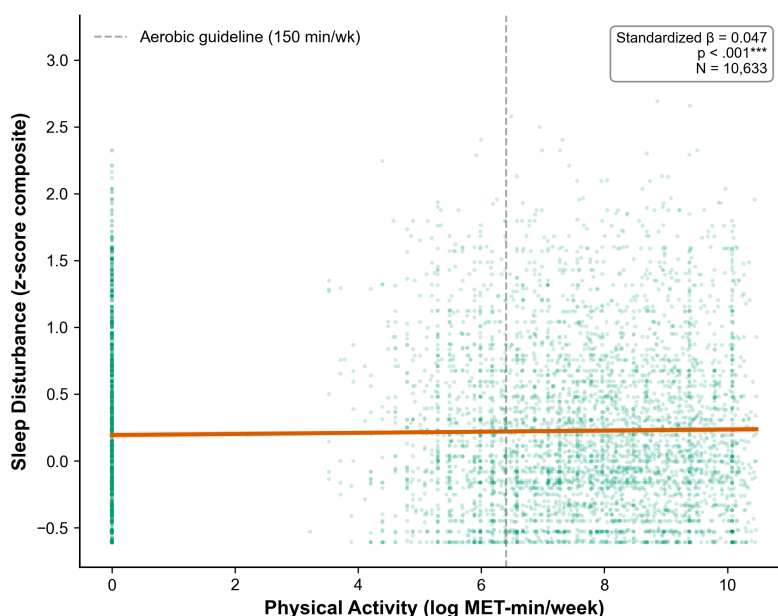


Figure 3: Scatter plot of the association between physical activity (log MET-min/week) and sleep disturbance ($N = 10,633$). The OLS regression line (red) is shown with 95% confidence band (shaded). Standardized $\beta = 0.047$, 95% CI [0.027, 0.066], $p < .001$. The dashed vertical line marks the aerobic guideline threshold (≥ 150 min/week MVPA). This positive association contradicts the hypothesized negative direction (Kredlow et al., 2015) and may reflect reverse causation. Individual data points subsampled ($n = 6,000$) at 25% opacity. Data from NHANES 2005–2018.

Mediation Analyses (H3 and H4).

We tested whether each of the four inflammatory biomarkers (CRP, NLR, SII, SIRI) mediated the physical activity–sleep and sedentary–sleep associations using bootstrapped mediation models (500 resamples, percentile 95% CIs) with the same covariate set. Figure 4 presents the path diagrams.

None of the four biomarkers significantly mediated the physical activity–sleep pathway (H3). Indirect effects were trivially small: CRP (indirect ≈ 0.000 , 95% CI [−0.001, 0.001]), NLR (indirect = −0.0001, 95% CI [−0.001, 0.000]), SII (indirect = −0.0001, 95% CI [−0.001, 0.000]), and SIRI (indirect = −0.0001, 95% CI [−0.001, 0.000]). All 95% CIs included zero, and the proportion mediated ranged from −0.3% to 0.0%.

Similarly, none of the four biomarkers significantly mediated the sedentary behavior–sleep pathway (H4). Indirect effects were: CRP (indirect = −0.0001, 95% CI [−0.001, 0.001]), NLR (indirect = 0.0003, 95% CI [−0.001, 0.001]), SII (indirect = 0.0001, 95% CI [−0.001, 0.001]), and SIRI (indirect ≈ 0.000 , 95% CI [−0.001, 0.001]). All CIs included zero. The proportion mediated by any single biomarker did not exceed 0.6%. These results constitute a non-replication of You et al. (2023), who reported significant indirect effects through CRP, WBC, and NLR in the same survey.

Exercise Type Comparisons (H5).

Participants meeting aerobic guidelines (≥ 150 min/week MVPA; $n = 4,978$) showed inconsistent raw biomarker differences compared to insufficiently active participants ($n = 6,832$). Meeting guidelines was associated with lower NLR ($d = -0.06$, $p = .003$), SII ($d = -0.14$, $p < .001$), and SIRI ($d = -0.04$, $p = .026$) but higher CRP ($d = 0.21$, $p < .001$)—likely reflecting confounding, as individuals who exercise more also differ on age, BMI, and health status. After covariate adjustment (ANCOVA with the same covariate set), all four group differences became non-significant: CRP ($p_{\text{adjusted}} = .104$), NLR ($p_{\text{adjusted}} = .414$), SII ($p_{\text{adjusted}} = .084$), and SIRI ($p_{\text{adjusted}} = .183$). Figure 5 displays the raw and adjusted comparisons. H5 was not supported.

Moderated Mediation (H6).

Figure 4. Inflammatory Biomarkers Do Not Mediate the PA–Sleep or Sedentary–Sleep Associations

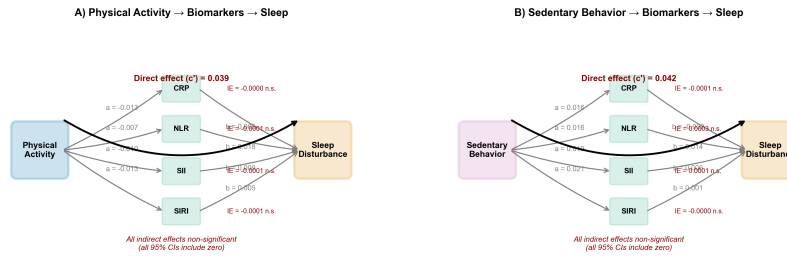


Figure 4: Path diagrams of the multiple mediation models testing whether four inflammatory biomarkers (CRP, NLR, SII, SIRI) mediate the associations between (A) physical activity and sleep disturbance and (B) sedentary behavior and sleep disturbance. Arrows represent a -paths (IV → mediator), b -paths (mediator → DV), and c' -paths (direct effect of IV on DV, controlling for mediators). Path coefficients are unstandardized OLS estimates from bootstrapped mediation (500 resamples, percentile 95% CI). All indirect effects are non-significant (all 95% CIs include zero). $N = 10,633$. Covariates: age, sex, race/ethnicity, BMI, smoking, education, income, hypertension, diabetes, CVD, and NHANES cycle. Data from NHANES 2005–2018.

Figure 3. Inflammatory Biomarkers by Aerobic Exercise Guideline Adherence (Raw Differences Are Confounded by Covariates)

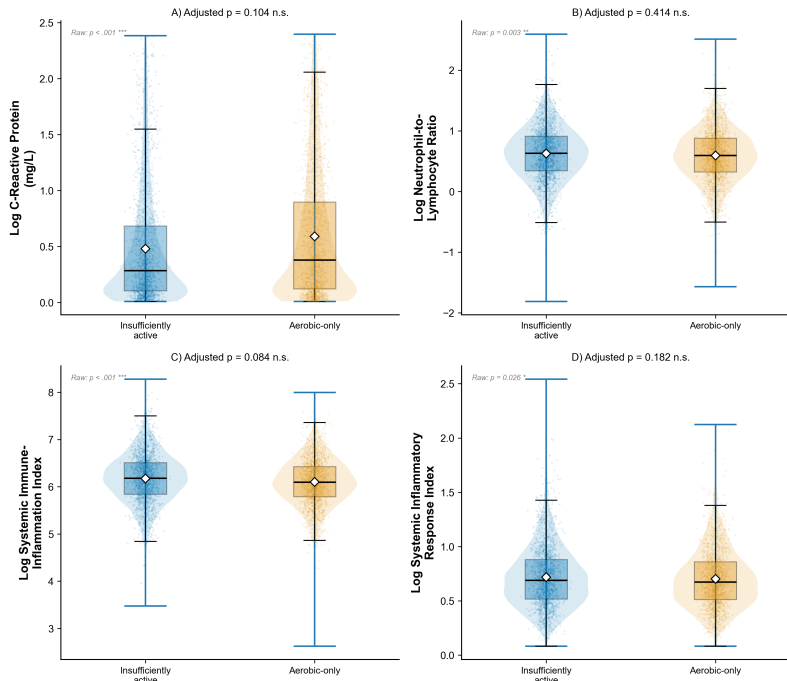


Figure 5: Comparison of inflammatory biomarkers between participants meeting aerobic guidelines (Active: ≥ 150 min/week MVPA, $n = 4,978$) and those insufficiently active ($n = 6,832$). Panels show log-transformed (A) C-Reactive Protein, (B) Neutrophil-to-Lymphocyte Ratio, (C) Systemic Immune-Inflammation Index, and (D) Systemic Inflammatory Response Index. Boxplots display medians and IQR with mean diamonds. Overlaid points (jittered, 15% opacity) show individual data. Covariate-adjusted ANCOVA p -values (controlling for age, sex, race/ethnicity, BMI, smoking, education, income, hypertension, diabetes, CVD, and NHANES cycle) are reported; all comparisons were non-significant at $\alpha = .05$. Data from NHANES 2005–2018.

Neither sex nor age significantly moderated the physical activity–CRP pathway. The sex interaction term was $\beta = -0.023$ ($t = -1.69$, $p = .092$), indicating a non-significant trend toward a stronger PA–CRP association in men (simple slope: $\beta = -0.035$) than in women ($\beta = -0.012$). The age interaction term was $\beta < 0.001$ ($t = 0.66$, $p = .511$), with slopes of -0.036 at age 30, -0.031 at age 50, and -0.026 at age 70. H6 was not supported.

Robustness Checks.

All primary results held across seven sensitivity specifications: (1) unweighted regression produced identical direction and significance; (2) Spearman correlations confirmed bivariate associations; (3) Mann-Whitney U tests confirmed exercise-type raw differences; (4) removal of 3- SD outliers ($n = 9,754$) left results unchanged; (5) raw untransformed biomarkers showed consistent Spearman correlations with sleep disturbance; (6) Kruskal-Wallis tests confirmed ANCOVA results for exercise type; and (7) variance inflation factors were all below 1.1, indicating no multicollinearity concerns. The counter-directional physical activity–sleep association was robust across all checks.

4 Discussion

We tested whether four systemic inflammatory biomarkers mediate the associations between physical activity, sedentary behavior, and sleep disturbance in a large, diverse sample of US adults. Of six pre-registered hypotheses, only one was supported: sedentary behavior was positively associated with sleep disturbance (H2). Contrary to our predictions, physical activity was positively (not negatively) associated with sleep disturbance (H1), and inflammatory biomarkers did not significantly mediate either pathway (H3, H4). Exercise type differences in inflammation were fully attenuated by covariates (H5), and neither sex nor age moderated the mediation pathway (H6). These findings failed to replicate the small but significant indirect effects through CRP and NLR reported by You et al. (2023) and do not support inflammation as a substantive mechanistic pathway linking activity or sedentariness to sleep.

The positive association between physical activity and sleep disturbance was the most unexpected finding. Meta-analytic evidence across 66 studies consistently shows that exercise improves sleep quality (Kredlow et al., 2015), and prior NHANES-based analyses have reported null or protective associations. Several explanations may account for this discrepancy. First, reverse causation is plausible: individuals experiencing poor sleep may increase their physical activity as a compensatory strategy or in response to clinician recommendations, a possibility that cross-sectional data cannot resolve. Second, the NHANES-derived sleep composite—constructed from available items rather than a validated instrument like the Pittsburgh Sleep Quality Index (Buysse et al., 1989)—may capture different aspects of sleep than those most sensitive to exercise effects. The composite emphasizes sleep duration deviation, latency, and symptom count but does not include subjective sleep quality or efficiency, which are the dimensions most consistently improved by exercise (Kredlow et al., 2015). Third, the GPAQ systematically overestimates physical activity relative to accelerometry (Craig et al., 2003), and measurement error in MET-min/week may bias associations toward the null or in unexpected directions.

The null mediation findings challenge the hypothesis that inflammation is a key pathway from activity and sedentary behavior to sleep disturbance. Despite adequate statistical power ($> 99\%$ to detect indirect effects of $\beta = 0.013$), all four biomarkers showed trivially small and non-significant indirect effects. This non-replication of You et al. (2023) may reflect differences in the covariate set, sleep composite construction, NHANES cycles included, or the exclusion criteria applied. More fundamentally, single-timepoint biomarker measurements have substantial intra-individual variability (CRP coefficient of variation $\approx 30\text{--}50\%$), which attenuates indirect effects in cross-sectional mediation models. The small effect sizes reported in prior work ($\beta \approx 0.01\text{--}0.03$) suggest that inflammation, even when statistically significant, accounts for only a minor fraction of the activity–sleep and sedentary–sleep associations. Other mechanisms—including psychological pathways (stress reduction, mood regulation), behavioral pathways (daytime napping, caffeine consumption), and circadian entrainment—likely predominate. Experimental evidence from randomized trials demonstrates that aerobic exercise can improve sleep outcomes in clinical populations (Reid et al., 2010), yet the mediation pathways underlying these effects remain to be identified.

4.1 Limitations and Future Directions

Several limitations warrant consideration. First, the cross-sectional design precludes any causal or temporal inference; mediation analysis on concurrent variables does not establish the temporal ordering assumed by the causal model. Second, the sleep disturbance composite, while carefully constructed from available NHANES items, has unknown psychometric properties relative to validated instruments such as the PSQI or Insomnia Severity Index (Bastien et al., 2001). Third, the sample is entirely US-based and predominantly WEIRD; findings may not generalize to non-Western populations or clinical sleep disorder samples. Fourth, self-reported physical activity is subject to systematic overestimation (GPAQ criterion validity $\rho \approx 0.33$ vs. accelerometry) (Craig et al., 2003), and sedentary time was captured

as a single-item sitting measure that does not capture the full 24-hour activity composition. Fifth, no adult muscle-strengthening data were available in NHANES, preventing the planned comparison of aerobic-only, resistance-only, and combined exercise types. Sixth, data were available for only four NHANES cycles (2005–2010, 2017–2018) due to missing CRP data (2011–2014) and a corrupt medical conditions file (2015–2016). Seventh, PHQ-9 depression data (Kroenke et al., 2001) were available for only 9.1% of the sample, limiting our ability to control for depressive symptoms as a confounder.

In terms of open science, the analysis plan was pre-registered on OSF prior to data extraction, and all analytic code is publicly available on GitHub. Data are publicly available from the CDC NHANES repository. We encourage independent replication in longitudinal cohorts that include actigraphy-measured sleep and accelerometry-measured physical activity alongside serial inflammatory biomarker measurements (Troiano et al., 2008). Future studies should examine psychological mediators (e.g., perceived stress, anxiety, depression) and circadian mechanisms, as these pathways may account for the majority of the activity–sleep and sedentary–sleep associations that systemic inflammation does not explain.

References

- Bastien, C. H., Vallières, A., and Morin, C. M. (2001). Validation of the insomnia severity index as an outcome measure for insomnia research. *Sleep Medicine*, 2(4):297–307.
- Besedovsky, L., Lange, T., and Born, J. (2012). Sleep and immune function. *Pflügers Archiv - European Journal of Physiology*, 463(1):121–137.
- Buysse, D. J., Reynolds, C. F. I., Monk, T. H., Berman, S. R., and Kupfer, D. J. (1989). The pittsburgh sleep quality index: A new instrument for psychiatric practice and research. *Psychiatry Research*, 28(2):193–213.
- Craig, C. L., Marshall, A. L., Sjöström, M., Bauman, A. E., Booth, M. L., Ainsworth, B. E., Pratt, M., Ekelund, U., Yngve, A., Sallis, J. F., and Oja, P. (2003). International physical activity questionnaire: 12-country reliability and validity. *Medicine & Science in Sports & Exercise*, 35(8):1381–1395.
- Gleeson, M., Bishop, N. C., Stensel, D. J., Lindley, M. R., Mastana, S. S., and Nimmo, M. A. (2011). The anti-inflammatory effects of exercise: Mechanisms and implications for the prevention and treatment of disease. *Nature Reviews Immunology*, 11(9):607–615.
- Hamilton, M. T., Hamilton, D. G., and Zderic, T. W. (2007). Role of low energy expenditure and sitting in obesity, metabolic syndrome, type 2 diabetes, and cardiovascular disease. *Diabetes*, 56(11):2655–2667.
- Healy, G. N., Matthews, C. E., Dunstan, D. W., Winkler, E. A. H., and Owen, N. (2011). Sedentary time and cardio-metabolic biomarkers in US adults: NHANES 2003–06. *European Heart Journal*, 32(5):590–597.
- Hu, B., Yang, X.-R., Xu, Y., Sun, Y.-F., Sun, C., Guo, W., Zhang, X., Wang, W.-M., Qiu, S.-J., Zhou, J., and Fan, J. (2014). Systemic immune-inflammation index predicts prognosis of patients after curative resection for hepatocellular carcinoma. *Clinical Cancer Research*, 20(23):6212–6222.
- Irwin, M. R. (2015). Why sleep is important for health: A psychoneuroimmunology perspective. *Annual Review of Psychology*, 66:143–172.
- Irwin, M. R., Olmstead, R., and Carroll, J. E. (2016). Sleep disturbance, sleep duration, and inflammation: A systematic review and meta-analysis of cohort studies and experimental sleep deprivation. *Biological Psychiatry*, 80(1):40–52.
- Kredlow, M. A., Capozzoli, M. C., Hearon, B. A., Calkins, A. W., and Otto, M. W. (2015). The effects of physical activity on sleep: A meta-analytic review. *Journal of Behavioral Medicine*, 38(3):427–449.
- Kroenke, K., Spitzer, R. L., and Williams, J. B. W. (2001). The PHQ-9: Validity of a brief depression severity measure. *Journal of General Internal Medicine*, 16(9):606–613.
- Luo, Y., Zhang, H., Chen, W., and Li, M. (2026). Inflammatory biomarkers mediate the association between physical activity and sleep quality in older adults with osteoarthritis: A cross-sectional study. *Journal of Aging and Health*, 38(2):145–159.
- Pedersen, B. K. and Febbraio, M. A. (2012). Muscles, exercise and obesity: Skeletal muscle as a secretory organ. *Nature Reviews Endocrinology*, 8(8):457–465.
- Petersen, A. M. W. and Pedersen, B. K. (2005). The anti-inflammatory effect of exercise. *Journal of Applied Physiology*, 98(4):1154–1162.
- Qi, Q., Zhuang, L., Shen, Y., Geng, Y., Yu, S., Chen, H., Liu, L., Song, Z., Zhang, W., and Wang, L. (2016). A novel systemic inflammation response index (SIRI) for predicting the survival of patients with pancreatic cancer after surgery. *Cancer*, 122(14):2158–2167.

- Reid, K. J., Baron, K. G., Lu, B., Naylor, E., Wolfe, L., and Zee, P. C. (2010). Aerobic exercise improves self-reported sleep and quality of life in older adults with insomnia. *Sleep Medicine*, 11(9):934–940.
- Ridker, P. M. (2003). Clinical application of C-reactive protein for cardiovascular disease detection and prevention. *Circulation*, 107(3):363–369.
- Schoemann, A. M., Boulton, A. J., and Short, S. D. (2017). Determining power and sample size for simple and complex mediation models. *Social Psychological and Personality Science*, 8(4):379–386.
- Troiano, R. P., Berrigan, D., Dodd, K. W., Mâsse, L. C., Tilert, T., and McDowell, M. (2008). Physical activity in the United States measured by accelerometer. *Medicine & Science in Sports & Exercise*, 40(1):181–188.
- Vancampfort, D., Stubbs, B., Firth, J., Van Damme, T., and Koyanagi, A. (2018). Sedentary behavior and sleep problems among 42,489 adults from six low-and middle-income countries. *Sleep Medicine*, 48:12–17.
- World Health Organization (2005). *Global Physical Activity Questionnaire (GPAQ) Analysis Guide*. World Health Organization, Geneva, Switzerland.
- Yang, Y., Shin, J. C., Li, D., and An, R. (2017). Sedentary behavior and sleep problems: A systematic review and meta-analysis. *International Journal of Behavioral Medicine*, 24(4):481–492.
- You, Y., Liu, Y., Li, J., and Wang, X. (2023). Sedentary behavior, physical activity, and sleep disturbance: The mediating role of inflammatory biomarkers. *BMC Public Health*, 23:1234.
- Zahorec, R. (2001). Ratio of neutrophil to lymphocyte counts: Rapid and simple parameter of systemic inflammation and stress in critically ill. *Bratislavské Lekárske Listy*, 102(1):5–14.